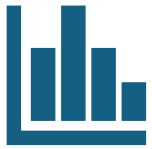
DATA 5300 Project 1

United Airlines Flight Delays Analysis

Agenda



Data and methodology



Findings of Time of Day and Time of Year



Findings of Weather-related factors



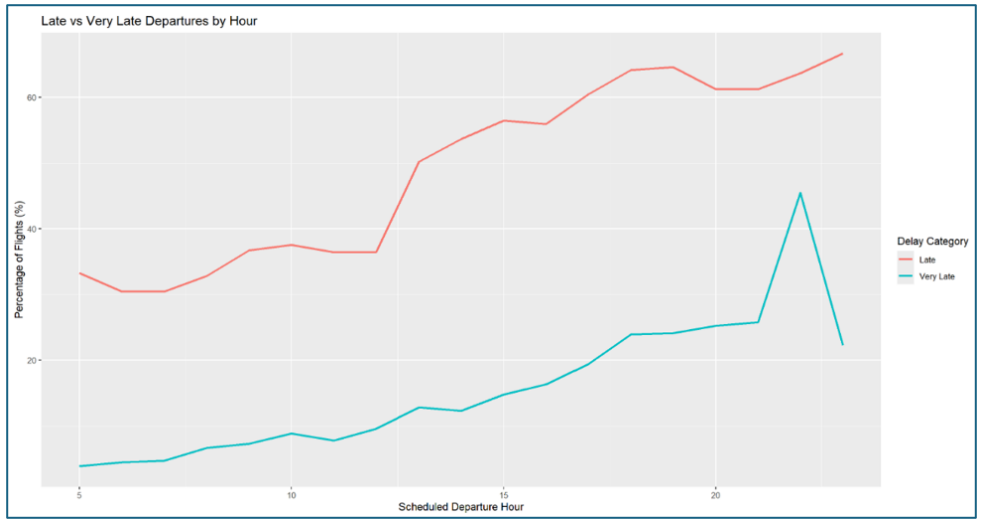
Conclusion

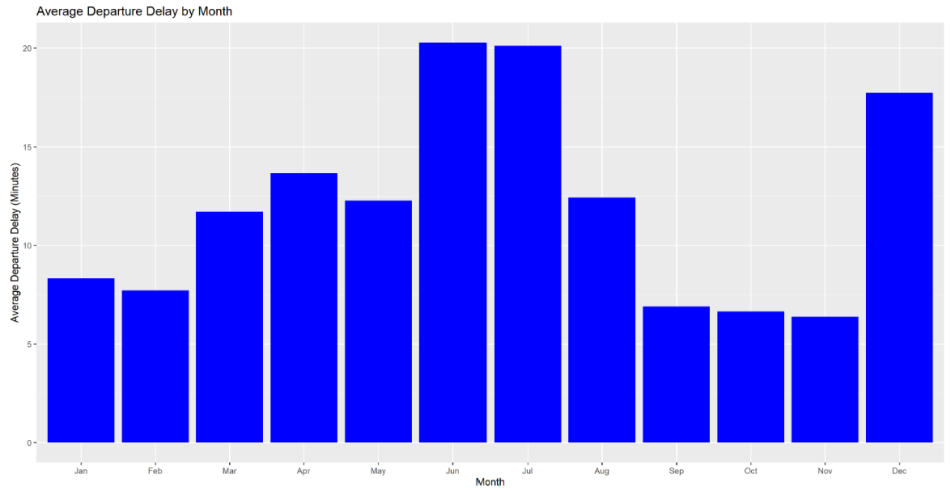
Data and Methodology

- Used *flights* and *weather* datasets from *nycflights13* library in R.
  - flights: flights data for NY area airports in 2013
  - weather: hourly weather data
- Data Preparation
  - Filtering and merging
    - United Airlines flights
    - Merged with hourly weather dataset
  - Handling missing data
    - Removed observations: 686 missing departure delays
    - 57,979 flights observations considered for analysis
- Data Augmentation
  - Late category → flights delayed by more than 0 minutes
  - Very Late category → flights delayed by more than 30 minutes

Findings

- Time of Day
  - Early morning: shorter average delays
  - Evening: longer delays
  - Difference in average delay: 18.16 minutes, Statistically significant
- Time of Year
  - June, July and December experienced notably longest average delays.



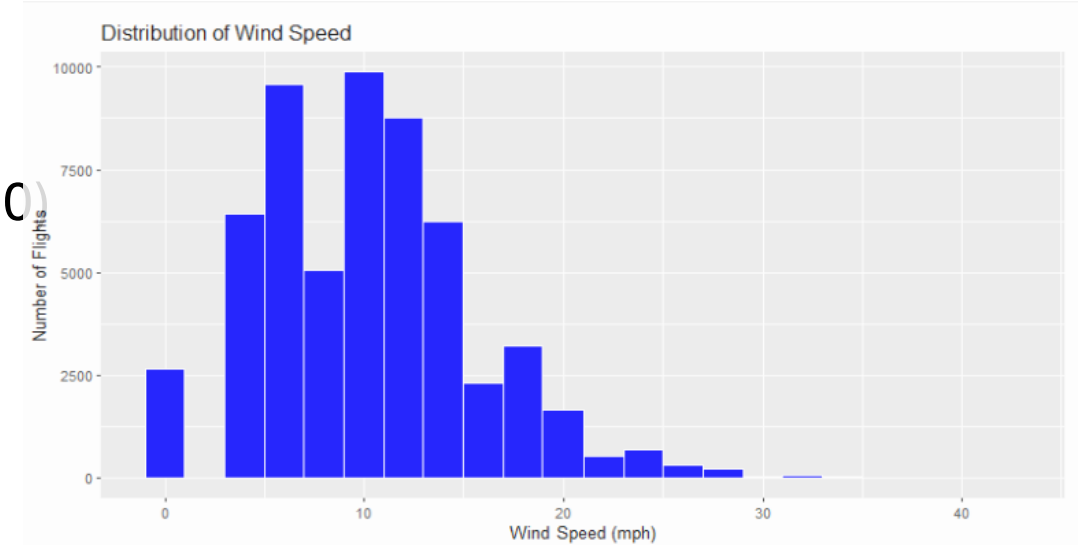


Findings

- Temperature
  - Cold (< 40F) vs Moderate (40F–80F) vs Hot (>80)
  - Statistical evidence hot vs moderate
    - 12.93 percentage points higher late departures
    - 6.64 percentage points higher very late departures
- Wind Speed
  - Most flights departed under low to moderate wind conditions
  - Difference between low and higher wind conditions were relatively small

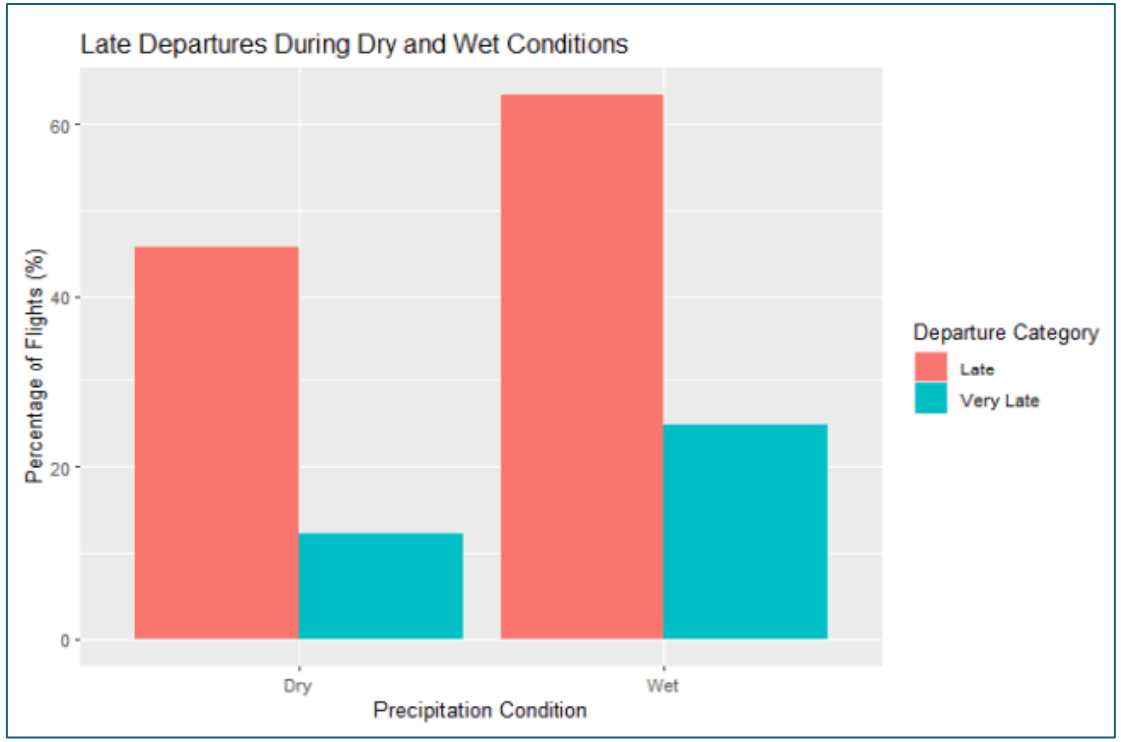
Departure Delays by Temperature Category

Temperature Category	Total Flights	Mean Delay (min)	Median Delay (min)	Late (%)	Very Late (%)
Cold	13033	9.94	0	47.43	11.09
Hot	7371	20.27	3	57.62	19.25
Moderate	37275	11.23	0	44.69	12.61



Findings

- Precipitation
  - Compared Dry vs Wet departures
  - Significant difference in late departure
    - 17.7 percentage points higher late
    - 12.55 percentage points higher very late
- Visibility
  - Compared normal vs reduced visibility
  - Statistically significant difference between normal and reduced visibility.
    - Both late and very late had similar difference of 5.23 and 5.24 percentage points, respectively.



Summary of findings

- Most notable difference in departure delays
  - Time of Day: Evening flights averaged 18.16 minutes more delays than morning
  - Precipitation: Wet weather had 17.7 percentage points higher late rate than dry conditions
  - Temperature: Hot temperature had 12.93 percentage points higher late rate than moderate temperature.
- Time of year, wind speed and visibility were also associated with departure delays.



Conclusion

- Opportunities for operational improvements
  - Investigate scheduling and staff allocations
  - Review wet weather procedures
  - Review hot temperature procedures
- Limitation
  - Analysis focused on limited data for one year and one geographical area.
- Enhancements
  - Expand dataset to include additional years, airports and airlines
  - Include destination information or other operational factors